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NUMERICAL OPTIMIZATION

Optimal Scheduling of Household Appliances under Time-Variable Electricity Tariffs

Professor: Izudin Džafić

Student: Suzan Alsayek

1. INTRODUCTION

Smart energy is a desktop application developed in C++ with the natID framework. The project combines a graphical user interface with a mathematical optimization model. The program actively calculates when household appliances should operate during a 24 hour period. The user enters the technical and scheduling parameters, starts the optimization and receives an optimized schedule together with cost, savings, power and sensitivity information. The central idea is demand shifting. If an appliance can operate within a flexible time interval, the program tries to move its operation toward lower tariff hours. At the same time, all enabled appliances must receive their required operating duration and the sum of their power must remain within the contracted household capacity. This is linear programming applied to smart home energy management.

OBJECTIVES:

1. Model a complete 24-hour household scheduling problem with four controllable appliances.
2. Minimize the modeled electricity cost using hourly tariff values.
3. Respect appliance time windows, required operating duration and the household power limit.
4. Present the result through an understandable visual dashboard instead of only numerical solver output.
5. Provide input validation, a default scenario and a bilingual Bosnian/English interface

SOFTWARE STRUCTURES:

OptimizationProblem - Stores tariffs, appliance parameters, time windows and household capacity; validates the input.

OptimizationSolver - Builds and solves the linear program with an internal two phase simplex method.

OptimizationResult - Stores the schedule, cost, savings, load profile, shadow prices and solver status.

ViewOptimizationInput - Collects user input and exposes language, tariff and appliance controls.

ViewOptimizationResults - Draws the dashboard, tariff graph, schedule, power profile and optimization insights.

MainView - Connects the input model, solver, result object and both visual views

2. OPTIMIZATION MODEL

Decision variables: For each of the four appliances and each of the 24 hours, the model creates a continuous scheduling variable $x(i,t)$. The project contains $4 \times 24 = 96$ primary decision variables. A value of 0 means that the appliance is not operating during that hour, a value of 1 represents full modeled operation and a value between 0 and 1 is possible because the project uses a continuous linear programming relaxation.

Appliance power: The program derives the average modeled power from the entered energy demand and duration using $P = \text{Energy} / \text{Duration}$. For example, an appliance that requires 4 kWh over 2 hours is modeled with an average power of 2 kW.

MODEL CONSTRAINT:

Required duration: for every appliance, the sum of its hourly schedule variables must equal the entered operating duration.

Allowed time window: an appliance can use energy only between its earliest start and latest finishing boundary. Outside this interval, its variable is forced to zero.

Household capacity: in every hour, the sum of appliance powers multiplied by their schedule values must be less than or equal to the maximum household connection power.

Variable bounds: $0 \leq x(i,t) \leq 1$. Disabled appliances are forced to zero for the whole day.

Equality handling: The internal simplex implementation accepts inequalities in the form $Ax \leq b$. So each duration equality is converted into two inequalities one upper inequality and one negative lower inequality. Upper bounds are also represented as linear constraints which keeps the complete scheduling problem in a standard linear programming form.

SOLVER IMPLEMENTATION:

A two phase simplex algorithm is implemented directly inside OptimizationSolver. The application is a minimization problem while the internal simplex routine maximizes an objective. The code maximizes the negative cost and converts the result back to the true minimum.

Phase I handles cases in which the initial basic solution is not feasible

Phase II optimizes the real objective after feasibility is established. Before solving, the input is validated. The program checks that household capacity is positive, tariffs are non-negative, energy demand and duration are positive, time boundaries are valid, duration fits inside the available time window and the power of one appliance does not exceed the household connection capacity. Invalid data is rejected before the LP model is generated.

3. USER INTERFACE

INPUT PANEL

- i. Language selection: Bosnian or English.
- ii. Home and power: maximum household connection power in kW.
- iii. Tariff: tariff profile plus editable lower and higher price values in KM/kWh.
- iv. Appliances: washing machine, dishwasher, water heater and storage heater.
- v. For every appliance: enabled/disabled state, energy demand, duration, earliest start and latest finish.
- vi. Actions: Optimize, Clear and Example/default scenario.

Tariff profiles - The interface supports predefined tariff patterns. In the summer working day profile, the lower tariff is used from 00:00-08:00, 14:00-17:00 and 23:00-24:00. The winter profile uses 00:00-07:00, 13:00-16:00 and 22:00-24:00. The Sunday profile applies the lower tariff for all 24 hours. The tariff vector is generated from the selected profile before the optimization is solved.

RESULTS DASHBOARD

- i. Optimized cost and the non-optimized reference cost.
- i. Percentage and absolute savings produced by optimization.
- ii. Peak household power and total scheduled energy.
- iii. Electricity-price chart showing lower and higher tariff periods across 24 hours.
- iv. Appliance schedule showing allowed windows in light colors and optimized activity in stronger colors.
- v. Household power chart showing hourly aggregate load relative to the connection limit.
- vi. Optimization insights containing dual values, capacity status and information about binding constraints.

APPLIANCE TIMELINE - For every device, the light colored interval shows the period in which the user allows the device to operate. The stronger colored blocks show where the optimizer actually placed its operation. This makes the result understandable without reading the underlying LP variables. If a fractional value is present, the fraction is displayed directly in the block.

LANGUAGES - The application contains separate Bosnian and English translation resources. Changing the language stores the selected translation and restarts the application so that text,

controls and regional number formatting are recreated consistently. This approach was used to avoid problems with numeric controls when changing locale while the interface was already active. Labels, buttons, tariff names and validation messages are translated, while the results canvas also switches its displayed text according to the active language.

4. RESULTS

The optimized solution is compared with a simple non-optimized reference schedule. The baseline algorithm places each enabled appliance as early as possible inside its allowed interval while still respecting the available household capacity. The application then calculates the difference between baseline cost and optimized cost. This produces both absolute savings in KM and percentage savings, which are shown in the summary cards. After the simplex solution is obtained, the program recalculates the true objective value from the final schedule rather than relying only on the tableau value. It also derives the hourly total power profile, peak power, total scheduled energy and the hours in which the capacity constraint is binding. These values are stored in `OptimizationResult` and are then used by the dashboard.

The project estimates shadow prices by slightly perturbing model constraints and resolving the LP. Capacity shadow prices indicate how the optimal cost changes if the power limit is relaxed at a specific hour. Appliance shadow prices estimate the cost effect of increasing the required operating duration. If the peak load remains below the household limit, the interface reports that the capacity limit was not reached instead of treating this as an error.